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HTML CSS JavaScript Bootstrap PHP MySQL GenAI Tools

Code for Good: AI-Powered Web App Development

Live Online Classes via Zoom

Build powerful web apps with AI. Create solutions. Make a difference.

Collaborate. Innovate. Create Impact.

Technology for Good

Empowering Minds! Inspiring Generosity!

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Code for Good (aka Upskill)

AI-Powered Web Application Development

Syllabus and Weekly Planner (30 Weeks)

Course Overview

This 30-week program teaches full-stack web development with real-world engineering practices. Students build and deploy a **real web application** for a nonprofit, business, or LLC while learning frontend, backend, databases, security, DevOps, and GenAI tools.

By the end, students will develop a **fully deployed, production-style web application for a LLC or non-profit.**

Program Phases

- **Phase 1:** Foundations and AI Tools (Weeks 1–6)
 - **Phase 2:** Frontend Engineering (Weeks 7–12)
 - **Phase 3:** Backend + Database + Security (Weeks 13–20)
 - **Phase 4:** DevOps + GenAI + Deployment (Weeks 21–26)
 - **Phase 5:** Capstone Delivery (Weeks 27–30)
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GenAI Integration (Across all Phases)

Students will use AI tools such as ChatGPT and Claude for:

- Learning and Exploring
- Planning and Designing
- Creating and Debugging code
- Generating CRUD systems
- Designing database schemas
- Writing documentation
- Improving UI/UX design

- Deployment of the full-stack applications
- Creating the presentations
- Registering non-profits or LLCs

Weekly Planner

Phase 1 — Foundations

Week	Topic	Key Outcomes
1	Introduction to Code for Good, Setup (GitHub, Claude and VS Code)	Web ecosystem + social impact mission
2	HTML Basics, Prompt Engineering Basics	Page structure, semantic tags
3	HTML Forms, Structured Prompt Templates	Input elements, validation basics
4	CSS Basics and Layouts	Styling, colors, typography, Responsive design
5	Project Selection - Phase 1	Incorporating LLC or Non-Profit; Governance
6	Git & GitHub Basics	Repositories, commits, push/pull

■ Phase 2 — Frontend Engineering

Week	Topic	Key Outcomes
7	JavaScript Basics	Variables, loops, functions
8	DOM Manipulation	Interactive UI development
9	Events & Validation	Form handling, interactivity
10	jQuery Basics	DOM simplification
11	Bootstrap / UI Frameworks	Responsive UI components
12	Mini Project 1	Responsive landing page



Phase 3 — Backend + Database + Security

Week	Topic	Key Outcomes
13	PHP Basics	Server-side programming
14	PHP Forms & Sessions	State management
15	MySQL Basics	Tables, queries
16	Database Design	ER diagrams, normalization
17	CRUD Applications	Create, Read, Update, Delete
18	Authentication System	Login, logout, password hashing
19	Role-Based Access Control (RBAC)	Admin, staff, user roles
20	Logging & Auditing	Activity tracking system

Phase 4 — DevOps + GenAI + Deployment

Week	Topic	Key Outcomes
21	Web Security Basics	OWASP concepts, input validation
22	Secure PHP Coding	SQL injection & XSS prevention
23	Advanced Logging & Auditing	System activity tracking
24	GitHub Workflow	Branching, pull requests, code reviews
25	Deployment Basics	Hosting with cPanel / shared hosting
26	Cloud Deployment Intro	AWS / Render / hosting fundamentals



■ Phase 5 — Capstone Project (Code for Good)

Week	Topic	Key Outcomes
27	Project Selection and Organization Setup	Incorporate nonprofit / business / LLC idea
28	Full Stack Integration	Frontend + backend + database
29	Testing & Deployment	Security checks, bug fixes, live hosting
30	Final Demo Day 🎤	Presentation to industry panel



Required Project Features

Each final project must include:

Authentication & Authorization

- Login / Logout system
- Secure password hashing
- Role-Based Access Control (Admin / User / Staff)

Database Design

- ER diagrams
- Normalized schema (1NF–3NF basics)
- Relational database with foreign keys

Security

- SQL injection prevention
- XSS protection
- Input validation
- Secure session handling

Logging & Auditing

- User activity logs
- CRUD operation tracking
- Audit trail tables

Responsive Design

- Mobile-first UI
- Bootstrap or CSS frameworks

GitHub Workflow

- Branching strategy

- Pull requests
- Code reviews
- Proper README documentation

Deployment

- Live hosted web application
- Domain (optional)
- Production-ready configuration

Final Outcome

By the end of this course, students will:

- Build full-stack web applications
- Design secure systems with RBAC
- Work with databases professionally
- Use GitHub like real developers
- Deploy production applications
- Use AI tools for software development
- Deliver a real-world impact project

Code for Good Mission

Students will create technology that supports:

- Nonprofits
- Small businesses
- Community organizations
- Social impact initiatives

They will finish the course not just as learners, but as **junior full-stack developers ready for real-world projects.**